Pickett 800-T Log Log — Cheat Sheet

THE SCALES

Scale	Shows	Range	Read with
C / D	x — the basic log scales	1 – 10	each other
CI / DI	1/x (inverted; read right→left, often red)	10 – 1	C / D
CF / DF	πx (folded at π)	π – 10π	C / D
CIF	1/(πx) (inverted folded)	—	C / CF
A / B	x^2 (two cycles)	1 – 100	D / C
K	x^3 (three cycles)	1 – 1000	D
S	sin θ (cos θ on complementary numbers)	5.74° – 90°	C (slide)*
T	tan θ	5.71° – 45°	C (slide)*
ST	sin ≈ tan ≈ radians, small angles	0.573° – 5.73°	C (slide)*
LL1	e^x, x = 0.01 – 0.1	1.010 – 1.105	D
LL2	e^x, x = 0.1 – 1	1.105 – 2.718	D
LL3	e^x, x = 1 – 10	2.718–22,000	D

* Always read a scale against the C or D scale it sits beside: trig scales on the slide go with C; anything on the body goes with D. If your S/T/ST sit on the body, use D instead of C in the trig steps.

READING & THE DECIMAL POINT

- The rule gives **digits only**. 2.4, 24, 0.024 all sit at the same spot.
- Check the **value of the smallest division** in each stretch — it changes (finer between 1–2, coarser at 4–10).
- **Place the decimal by rough estimate:** 38.2 × 0.0517 ≈ 40 × 0.05 = 2 → digits 1975 → **1.975**.
- Or use powers of ten: write each number as d.ddd × 10^n; add exponents for ×, subtract for ÷; adjust by 1 if the slide went "off the right end."
- **Index** = the 1 at either end of C (left index or right index).

MULTIPLICATION a × b

1. Set the C index over **a** on D.
2. Move the hairline to **b** on C.
3. Read the answer on D under the hairline.

Off the end? Use the other C index (right index instead of left), or work on **CF/DF**: same steps using the CF index — it's near the middle, so the answer almost never falls off.

> **Ex:** 2.4 × 3.5 → left C index on 2.4 (D), hairline to 3.5 (C), read **8.4** on D.
> **Ex:** 6 × 7 → off the right with the left index; use the right C index on 6, hairline to 7 → 42 → **42**.

Three factors a × b × c (one setting)

Set **b on CI** over **a on D**; move hairline to **c on C**; read D.

DIVISION a ÷ b

1. Put the hairline on **a** on D.
2. Slide **b** on C under the hairline.
3. Read the answer on D under the C index.

> **Ex:** 8.4 ÷ 3.5 → 3.5 (C) over 8.4 (D); under C index read **2.4**.

Chained (a × b) ÷ c

Divide first: **c on C over a on D**; hairline to **b on C**; read D. Keep alternating ÷, × — each step only moves the hairline or the slide.

Proportion a / b = c / x

Set **a on C over b on D**. Now every C/D pair has the same ratio: find **c** on C, read **x** on D under it (or the reverse). Great for unit conversion and percentages.

RECIPROCALS & π

Want	Do
1/x	Hairline on x on C → read CI (or x on D → read DI)
πx	Hairline on x on D → read DF (C → CF)
x/π	Hairline on x on DF → read D
1/(πx)	Hairline on x on C → read CIF
a ÷ b via CI	C index over a on D, hairline to b on **CI**, read D (÷ becomes ×)
Circumference	Diameter on D → read πd on DF

SQUARES & SQUARE ROOTS (A, B)

x^2: hairline on x on D → read A. (Slide: C → B.)

$\sqrt{x}$: hairline on x on A → read D. Pick the correct **half** of A:

- Mark off digits in **pairs from the decimal point** (both directions).
- Leftmost non-zero group has **1 digit → left half** (1–10); **2 digits → right half** (10–100).
- Result has one digit before the point per group left of the point.

> **Ex:** √250 → "2 50" → leading group 1 digit → left half, 2.5 → D: 1581 → **15.81**.
> **Ex:** √2500 → "25 00" → right half, 25 → D: **50.0**.
> **Ex:** √0.0036 → "0.00 36" → right half 36 → 6 → **0.06**.

Area of circle: d on D → d^2 on A, then × 0.7854 (= π/4).

$1/x^2$: x on CI → read B.

CUBES & CUBE ROOTS (K)

x^3: hairline on x on D → read K.

$\sqrt[3]{x}$: hairline on x on K → read D. Group digits in **threes from the decimal point**; leading group of 1, 2, or 3 digits → left, middle, or right third of K.

> **Ex:** ∛50 → "50" (2 digits) → middle third, 50 → D: **3.684**.
> **Ex:** ∛5000 → "5 000" → left third, 5 → D: 1710 → **17.10**.

$x^{3/2}$: x on A → read K. **$x^{2/3}$**: x on K → read A.

HANDY CONSTANTS

π = 3.1416	e = 2.7183	π/4 = 0.7854
ln 10 = 2.3026	$\log_{10}e$ = 0.4343	√2 = 1.4142
1 rad = 57.296°	1° = 0.017453 rad	√3 = 1.7321
g = 9.807 m/s² = 32.17 ft/s²	1 in = 2.54 cm	1 lb = 0.4536 kg

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SINE, COSINE, TANGENT

Want	Set	Read	Value range
sin θ, 5.74°–90°	θ on S	C	0.1 – 1.0
sin θ, 0.573°–5.73°	θ on ST	C	0.01 – 0.1
cos θ	(90° – θ) on S (or red cos numbers)	C	= sin(90°–θ)
tan θ, 5.71°–45°	θ on T	C	0.1 – 1.0
tan θ, 45°–84.3°	(90° – θ) on T	**CI**	1.0 – 10
tan θ, 84.3°–89.4°	(90° – θ) on ST	**CI**	10 – 100
cot θ	θ on T	**CI**	= 1/tan θ
csc θ, sec θ	θ on S (sec: 90°–θ)	**CI**	1/sin, 1/cos
θ from value	value on C	S / T / ST	reverse lookup

Angles < 0.573°: sin θ ≈ tan θ ≈ θ (radians) = θ° × 0.017453. For angles > 90°, reduce first: sin(180°–θ) = sin θ, cos(180°–θ) = –cos θ, etc.

Ex: sin 30° → 30 on S → **0.500** on C. tan 20° → 20 on T → **0.364**.
Ex: tan 70° → 20 on T → read CI: **2.747**. sin 2° → 2 on ST → **0.0349**.

DEGREES ↔ RADIANS

Small angles: θ° on ST → radians on C (0.01–0.1). 1° → 0.01745.

Any angle: proportion — set **57.3 on C over a D index** (left or right, whichever keeps your number on the rule). Then degrees on C ↔ radians on D.

Ex: 40° → 0.698 rad. 2.5 rad → 143.2°.

MULTIPLY / DIVIDE BY A TRIG VALUE

a × sin θ: C index over a on D; hairline to θ on S; read D.

a ÷ sin θ: hairline on a on D; slide θ (on S) under hairline; read D under C index.

Same pattern with T and ST. Use the other index if it runs off.

Ex: 12 × sin 35° → C index on 12, hairline to 35 on S → **6.88**.

TRIANGLES

Law of sines a / sin A = b / sin B = c / sin C

1. Set angle **A on S** over side **a on D**.
2. Hairline to angle **B on S** → side **b on D**. (Also works in reverse: find a side on D, read its angle on S.)

Ex: a = 10, A = 40°, B = 65° → 40 (S) over 10 (D); hairline to 65 (S) → b = **14.10**.

Right triangle (legs a, b; hypotenuse c)

- **Angle:** tan θ = a/b (a < b). Set right C index over b on D; hairline to a on D; read θ on T.
- **Hypotenuse:** c = a / sin θ. Leave hairline on a (D); slide θ on S under hairline; read c on D at C index.
- Also handy: polar ↔ rectangular (x = r cos θ, y = r sin θ) — same steps.

Ex: legs 3, 4 → tan θ = 0.75 → θ = **36.87°**; c = 3 / sin 36.87° = **5.00**.

LOG LOG BASICS (LL scales read on D)

Each LL value sits **directly over ln(value) on D**. The scale tells you where the decimal point of the exponent goes:

Scale	Values	x in e^x	D reading 2.3 means
LL1	1.010 – 1.105	0.01 – 0.1	$e^{0.023} = 1.0233$
LL2	1.105 – 2.718	0.1 – 1	$e^{0.23} = 1.2586$
LL3	2.718–22,026	1 – 10	$e^{2.3} = 9.974$

Key fact: any LL value raised to the **10th power** sits on the **next scale up at the same hairline** (LL1 → LL2 → LL3). Example: 1.06 on LL1 is directly over $1.06^{10} = 1.791$ on LL2.

e^x and ln x

e^x: hairline on x on D → read LL scale chosen by the size of x.

ln y: hairline on y on its LL scale → read D; decimal point from the scale (table above).

$\log_{10}$ y (no L scale): $\log_{10} y = \ln y \times 0.4343$. **$\log_b$ y** = ln y ÷ ln b (two LL lookups + a C/D division).

Ex: $e^{0.035}$ → 3.5 on D → LL1: **1.0356**. $e^{2.3}$ → LL3: **9.97**. ln 7.5 → 7.5 on LL3 → D 2015 → **2.015**.

POWERS y^p

1. Set the C index over **y** on its LL scale.
2. Move the hairline to **p** on C.
3. Read y^p on an LL scale under the hairline:

- **Left C index used** (p read as 1–10) → answer on the **same** LL scale as y.
- **Right C index used** (because the left went off the end) → answer on the **next scale up**.
- If p is 10–100 → move one more scale up; if 0.1–1 → one scale down. (Each ×10 in p = one scale.)
- Sanity check: ln(answer) should equal p × ln y — the scale must cover that range.

Ex: $3^{4.2}$ → left index on 3 (LL3), hairline 4.2 → LL3: **100.9**.
Ex: $1.05^{2.5}$ → left index runs off; right index on 1.05 (LL1), hairline 2.5 → LL2: **1.1297**.
Ex: $1000 at 6% for 10 yr → 1.06^{10}: hairline on 1.06 (LL1) → straight to LL2: **1.791** → $1791.

ROOTS $\sqrt[n]{y} = y^{1/n}$

1. Hairline on **y** on its LL scale.
2. Slide **n** on C under the hairline.
3. Read the root on LL under the C index (left index → same scale; right index → **one scale down**).

Ex: $\sqrt[3]{50}$ → hairline 50 (LL3), slide 3 on C under it → under left index, LL3: **3.684**.
Ex: $\sqrt[5]{2}$ → hairline 2 (LL2), slide 5 under it → under left index, LL2: **1.1487**.
Ex: $\sqrt[3]{1.2}$ → left index runs off; read under right index, one scale down (LL1): **1.0627**.

Solve for an exponent ($y^p = z$): p = ln z ÷ ln y — read both on D, divide on C/D.

VALUES BELOW 1 & NEGATIVE EXPONENTS

This rule has no LL0/LL00 scales, so use reciprocals:

- **e^{-x}** = 1 / e^x: find e^x on LL, then 1/that via C → CI.
- **y^{-p}** = 1 / y^p. **$(0.5)^p$** = 1 / 2^p.
- **ln y for y < 1:** ln y = –ln(1/y).

Ex: $e^{-0.7}$ → $e^{0.7}$ = 2.014 (LL2) → 1/2.014 = **0.497**.